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Aldrin Joan Pandian W

Programming Languages: Java, C++, Python
Mobile Development: Android Development, Flutter, React Native
Machine Learning & AI: Natural Language Processing, Machine Learning, Generative AI (GPT-5, LLM fine-tuning), Multi-Agent Systems, Prompt Engineering, Model Context Protocol Server.
Web Development: Web Development (React JS, Next JS), API Integration, JavaScript, WebSocket.
Cloud & DevOps: Azure Deployment, Cloud-based AI applications
Tools & Frameworks: LangChain, LangGraph, LangSmith, YOLO model, Responsive Design, Real-time Data Handling, File Processing, Automation, FastMCP, IDE Extensions.
Certification:

- EC Council Certified Ethical Hacking V12 – EC Council (2023-27)
- AI Engineering – IBM

EDUCATION

Board	Tenure	Educational Institution	CGPA/Percentage
B. Tech (CSE) AI&ML	September 4, 2023 - Ongoing	VIT Bhopal University, Bhopal	8.89/10
Class XII	May 10, 2023	Carmel Garden Matric Hr Sec School, Coimbatore	75.0%
Class X	May 28, 2021	Carmel Garden Matric Hr Sec School, Coimbatore	100%

ACADEMIC PROJECTS

End-to-end Developer

- **paperstack — MCP Backend for arXiv Research (2026) (477 Total Downloads from PyPi)**
  - **Description**: Built a production-grade Model Context Protocol (MCP) server for end-to-end arXiv research workflows, enabling automated paper retrieval, PDF parsing, and LLM-ready context generation for AI agents and developer tools.
    - End-to-end pipeline: arXiv search → PDF fetch/cache → text extraction → token-aware chunking
    - Exposes 10+ MCP tools for agent workflows (search, parsing, citation graph, semantic indexing, paper comparison)
    - Advanced research intelligence layer: contribution extraction, citation networks, reproducibility scoring, implementation diffing
    - Built developer tooling layer for code/dataset link extraction and reproducibility audits
    - Supports autonomous workflows: reading lists, topic tracking, and audience-specific explanations
    - Robust backend design with caching, retry logic, rate limiting, and async I/O for high reliability
    - Token-aware context builder optimized for LLM pipelines and agent-based systems
    - **Tech:** Python, FastAPI-style MCP server, AsyncIO, httpx, PyMuPDF, Pydantic, SQLite, REST APIs, Sentence Transformers
    - **Role:** Sole Developer (Architecture → Backend Systems → Packaging & Deployment)
  - **Publishing & Distribution**
    - Packaged and published as a Python library on PyPI (paperstack-mcp).
      - **https://pypi.org/project/paperstack-mcp**
    - CLI + API + MCP server integration for VS Code and agent frameworks
      - **https://github.com/Aldrin-Joan/paperstack**
    - Production-ready packaging with pyproject.toml, dependency management, and test suite

End-to-end Developer

- **GraviGauge — VS Code Extension (2026) (1142 Downloads from OpenVSX Market)**
  - **Description:** Built a production-ready VS Code extension to monitor real-time AI model quotas from Antigravity, achieving 1,142+ downloads. Reverse-engineered internal APIs for secure, zero-config integration.
    - Real-time quota display in status bar with health indicators
    - Auto-detects running instance & handles session-based authentication
    - Smart polling + manual refresh with minimal performance overhead
    - Robust error handling and detailed model insights (Gemini, Claude, GPT)
    - **Tech**: TypeScript, Node.js, VS Code API, REST APIs

	▪ **Role**: Sole Developer (Architecture → Deployment) - **Publishing:** • Published on Microsoft VS Code Marketplace ○ **https://marketplace.visualstudio.com/items?itemName=AldrinJoanPandianW.gravigauge** • Published on Open VSX Registry ○ **https://open-vsx.org/extension/AldrinJoanPandianW/gravigauge** • Complete extension packaging, semantic versioning, and release management
End-to-end Developer	▪ **Bollywood Bias Buster (June 30, 2025 – June 19, 2025)** - **Description**: Detect, quantify, and remediate gender bias in Bollywood media. Classifies script dialogues by stereotype, rewrites biased text, visualizes bias trends, and generates structured PDF/CSV reports. Extended to posters (LLaVA-1.5) and trailers (emotion–gender plots) with Wikipedia-based linguistic and network analytics. - **Features**: ▪ Data Cleaning — Extract gender-tagged dialogues from scripts (PDF → CSV) using NLP-based character and gender identification ▪ Classification — Detect gender stereotypes with local Mistral-7B (GGUF) ▪ Visualization — Bias category distribution, gender-wise trends ▪ Rewrite — Stereotype-free rewriting using LLM guidance ▪ Report Generation — Downloadable PDF + CSV reports ▪ Poster Analysis — Visual bias detection with LLaVA-1.5 ▪ Trailer Analysis — Emotion–gender timeline plots ▪ Wikipedia Analytics — Adjective/verb usage, centrality, coreference, songs metadata - **Technology**: Streamlit, Hugging Face Transformers (Mistral-7B, LLaVA-1.5), PyMuPDF, Pandas, Matplotlib, Seaborn, NetworkX, NLP pipelines - **Team Project**: Individual - **Role:** End-to-end Developer — bias classification, multimodal integration, UI/UX, and report generation - **GitHub: https://github.com/Aldrin-Joan/Bolly_Bias_Streamlit_By_Aldrin_Joan_Pandian.git**
Android App & Large File Processing using AI & ML	▪ **Legal AI Android & Web App** (March 29, 2025 – Ongoing) - **Description**: Developed a cross-platform AI legal assistant for Android and web. Enabled users to post legal issues, get AI-generated insights, access legal documents, and receive real-time legal updates with misinformation checks. Included voice-based multilingual chatbot and clause analysis. - **Technology**: React.js, Flutter, Django, FastAPI, MongoDB, Pegasus Indian Legal, CUAD, SpaCy, NLTK, Google STT/TTS, TD-IDF + LDA, AWS, Firebase, LangGraph - **Team Project:** 4 Members - **Role**: Front End (ReactJS), Flutter App Developer and Large File Processing using AI & ML
Flutter Developer	▪ **RehabEase – AI-Powered Rehabilitation Platform** (Dec 20, 2024 – May 26, 2025) - **Description**: Designed a mobile and web app for guided rehab exercises. Integrated motion tracking, real-time feedback, and progress monitoring. - **Technology**: Python Flask, Dart, Flutter, Firebase and Computer Vision - **Team Project**: 3 Members - **Role**: Team Leader and Android and IOS App Developer
Front End	▪ **Deep Learning-Based Knee Osteoarthritis Diagnosis** (Jan 29, 2025 – April 14, 2025) - **Description**: Implemented CNN and ensemble models to classify OA severity from X-rays. Added Grad-CAM and a web interface for real-time diagnosis. - **Technology**: Python, CNN, ReactJS, Machine Learning - **Team Project**: 6 Members - **Role**: Front End Developer
Model Training	▪ **Heart Disease Prediction Using Machine Learning** (Sep 23, 2024 – Dec 18, 2024) - **Description**: Created a machine learning model to predict heart disease risk. Applied feature engineering, classification, and visualization techniques. - **Technology**: Python, Jupyter, Machine Learning - **Team Project**: 2 Members - **Role**: Team Leader and Model Training
Model Training	▪ **Hand Sign Language Translation Using CNN** (Oct 21, 2024 – Dec 18, 2024) - **Description**: Built a CNN-based model to convert hand signs to text and vice versa. Enhanced gesture recognition using computer vision for accessibility. - **Technology**: Python and CNN Model - **Team Project**: 5 members - **Role**: Model Training
Front End	▪ **AI-Powered Mentor-Mentee Platform** (March 14, 2024 – Oct 23, 2024)

	- **Description**: Developed a web app with AI-based matching to connect mentors and mentees. Integrated real-time chat, video calls, and calendar booking. - **Technology**: NextJS, Django, Firebase, MySQL, Python - **Team Project**: 6 Members - **Role**: Team Leader and Front-End Developer

INTERNSHIP	
Docu3C (Seattle, Washington, USA) (July 21, 2025 – Till Date)	**AI Engineer** • **Prompt Engineering & Optimization**: Designed and optimized effective prompts for Large Language Models (LLMs) including GPT-5 Family, Claude, and DeepSeek to drastically improve response accuracy, reliability, and contextual awareness. • **Multi-Agent System Development**: Architected and deployed intelligent, collaborative multi-agent workflows using LangChain, LangGraph, LangSmith, LiteLLM, and Temporal orchestration. • **AI Research & Development (R&D)**: Spearheaded R&D initiatives to evaluate emerging LLM capabilities and prototype experimental multi-agent applications for specialized domain tasks. • **Advanced NLP & Semantic Analysis**: Applied advanced Natural Language Processing techniques for text analysis, topic modeling, and semantic/vector search capabilities. • **Cloud-Based AI Deployment**: Developed, optimized, and deployed scalable AI applications on Microsoft Azure, ensuring high performance, efficient resource orchestration, and seamless system integration. • **Automated Workflows & Integrations**: Built automated, context-aware applications for real-time data handling, complex file processing, and deep integrations with Salesforce and ServiceNow. • **Agentic System Optimization**: Spearheaded the debugging, optimization, and error-handling of complex agentic frameworks to ensure robust system reliability in production environments. • **Protocol & API Expansion**: Upgraded system function-calling capabilities by building and integrating robust Model Context Protocol (MCP) Servers. • **Engineering & Automation:** Salesforce & ServiceNow Automation, Real-time Data Handling, File Processing Pipeline Development, Debugging & Error Handling. • **Hybrid Expert Reasoning Systems:** Architecting a system that combines deterministic legal heuristics with probabilistic Large Language Model (LLM) agents. • **Calibration Framework:** Developed a high-scale benchmarking framework to validate AI reasoning against "Ground Truth" expert data. • **End-to-End (E2E) Application Delivery:** Acted as the sole developer to architect, build, and deploy complete AI-driven applications from the ground up, managing the full software lifecycle from initial concept to final Azure deployment. **Skills** • **AI & Orchestration:** Prompt Engineering, Multi-Agent Systems, Multi-Model Agentic Systems, AI Research & Development (R&D), LangChain, LangGraph, LangSmith, LiteLLM, Temporal. • **Generative AI & LLMs:** GPT-5, Claude, DeepSeek, LLM Fine-tuning, Context Management, Function Calling, Model Context Protocol (MCP) Servers. • **Natural Language Processing:** TF-IDF, LDA, NMF, Semantic Search, Vector Search, Text Analysis. • **Cloud & Architecture:** Azure Deployment, Cloud-based AI Applications, Scalable Architecture, System Orchestration. • **End-to-End (E2E)** Application Development.
amasQIS.ai (Omen) (April 28, 2025 – October 31st, 2025)	• Worked as part of the Front-End team for the Personal Protective Equipment Compliance Project, contributing to webpage design and UI development. • Served as a Backend Developer in the Predictive Maintenance Project, integrating the YOLO model to meet custom requirements for defect detection. • Acted as Lead Flutter Developer for two projects — LLM Workspace Adviser and VLM Workspace Adviser. • Handled end-to-end mobile application development.
ApexPlanet Software Pvt. Ltd. (Oct 20, 2024 – Dec 4, 2024)	• Completed structured tasks involving webpage creation, responsive design, JavaScript interactivity, API integration, and localStorage. • Developed a dynamic to-do list, image gallery, and an E-commerce website as the final project. • Enhanced frontend development skills and implemented real-time data handling and performance optimization techniques.

CO-CURRICULARS	
Coding	▪ http://www.skillrack.com/profile/452987/63a1b8e881639ac762c3e1187e2cdf1f0f66308a ▪ 87600 Rank

EXTRA-CURRICULARS AND ACHIEVEMENTS	
Achievements	▪ Research Paper Presented, 3rd International Conference ICRP-2024 – ICRP (2024) ▪ Research Paper Presented, Third International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI 2025) ▪ Research Paper Presented, Third International Conference on Next Generation Computing Systems (ICNGCS'25) ▪ Attended, CSIR Sponsored National Seminar on Vehicular Cloud Model – CSIR (2024)
Responsibilities	▪ UGC NEP SAARTHI – Student Ambassador (Jan 02, 2025 – Till Date)
Extracurricular	▪ Completed 7 Grades in Keyboard – London College of Music (LCM) & Trinity College of Music

ADDITIONAL INFORMATION	
Research	▪ Pandian, W. Aldrin Joan, et al. "The Role of Artificial Intelligence in 6G Networks, Architecture, Protocol, Transmission, and Applications." *RFID, Microwave Circuit, and Wireless Power Transfer Enabling 5/6G Communication*. IGI Global Scientific Publishing, 2025. 115-154. ▪ Pandian, W. Aldrin Joan, I. Jasmine Selvakumari Jeya, and D. Lakshmi. "for Smart Grid Components: Architecture, Applications, Challenges, and Opportunities." *Renewable Power for Sustainable Growth: Proceedings of ICRP 2024, Volume 2* 2 (2026): 405. ▪ Aldrin Joan Pandian, W., et al. "AI-Driven Predictive Maintenance for Smart Grid Components: Architecture, Applications, Challenges, and Opportunities." *International Conference on Renewable Power*. Singapore: Springer Nature Singapore, 2024. ▪ J. S. Jeya I, A. J. Pandian W and P. K. Mishra, "A Next-Generation Computing Approach for Early and Accurate Lung Cancer Detection Using an Integrated Hybrid Deep Learning Framework," *2025 International Conference on Next Generation Computing Systems (ICNGCS)*, Coimbatore, India, 2025, pp. 1-8, doi: 10.1109/ICNGCS64900.2025.11183343. ▪ Jeya, Jasmine Selvakumari, J. Manikandan, and A. Sherin. "Optimizing Video Compression for Real-Time and Bandwidth-Constrained Applications in IoT Enabled Cyber-Physical Systems." *2025 3rd International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*. IEEE, 2025. ▪ J. S. Jeya I, N. A. More, B. Das, R. Gupta, V. K. Mishra and A. J. Pandian W, "IoT Integrated Smart Medication System using Machine Learning for Enhanced Rural Healthcare," *2025 6th International Conference on IoT Based Control Networks and Intelligent Systems (ICICNIS)*, Bengaluru, India, 2025, pp. 442-448, doi: 10.1109/ICICNIS66685.2025.11315818.
Hobbies	▪ Reading Japanese Mangas ▪ Playing Games ▪ Playing Basketball
Languages	▪ English, Tamil, Hindi, Japanese